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Roll No

EC-8004(1)-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Advanced Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Classify and explain different types of signals and systems.
b) Determine if the system described by the following input output equation are linear time invariant or not.
 - i) $y(n) = nx(n)$
 - ii) $y(n) = e^{x(n)}$
 - iii) $y(n) = x^2(n)$
 - iv) $y(n) = x(2n)$
2. State and prove differentiation, integration, convolution and frequency shifting property of Z-Transform.
3. a) Explain about Average spectrum representation of Infinite Energy Signals.
b) Discuss about response of Linear System to Random Signals.

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4. a) Explain radix-4 FFT algorithm in detail.
b) Explain the linear convolution using DFT.
5. Compute $N = 8$ point DFT using Radix-2 decimation in time FFT algorithm.
6. a) Explain about forward and backward linear prediction.
b) Explain about sampling rate conversion by a Rational factor I/D .
7. a) Determine the causal signal $x(n)$ whose Z-transform is
given by $X(z) = \frac{1 + z^{-1}}{1 - z^{-1} + 0.5z^{-2}}$
b) Discuss filter design and implementation for sampling rate conversion.
8. a) Differentiate between DTFT and DFT.
b) Explain Least Square Method.
c) Explain Power Spectrum.
